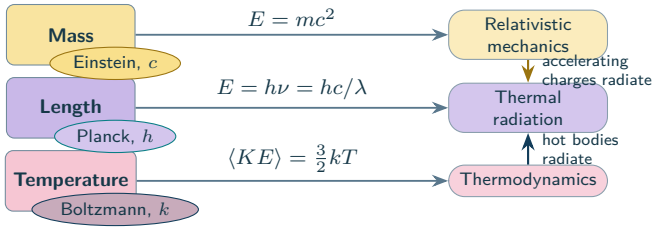
ENERGY ON ONE PAGE

A photon's journey: from $E = mc^2$ to warm air

Three constants, c, h and k, turn mass, frequency and temperature into energy along the way.

1 / THE RULE AND THE PICTURE

Each constant converts one kind of quantity into energy



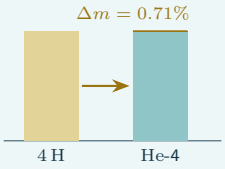
what 1 eV is, through each constant

mass, $m = E/c^2$	1.78×10^{-36} kg
frequency, $\nu = E/h$	2.42×10^{14} Hz
wavelength, $\lambda = hc/E$	1239.8 nm
temperature, $T = E/k$	11 605 K

Energy is the common currency, and the photon is the carrier that passes it from the Sun's nuclei to the air.

2 / NUCLEAR / BIRTH IN THE CORE, c

Four protons become helium and 0.7% of the mass becomes energy

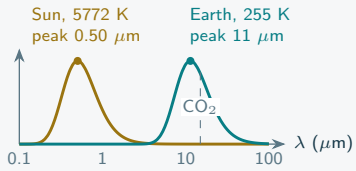


Four hydrogen atoms weigh 4.031300 u and one helium-4 atom 4.002603 u, so $\Delta m = 0.02870$ u and $E = \Delta m\, c^2 = 26.7$ MeV; the Sun's $L = 3.83 \times 10^{26}$ W costs $L/c^2 = 4.3 \times 10^{9}$ kg every second. The energy appears as MeV gamma rays, but at 1.5×10^{7} K the thermal scale is $kT = 1.3$ keV, and repeated absorption and re-emission shares each MeV among about 10^6 photons of roughly 1 eV over 10^4 to 10^5 years.

$$E = \Delta m\, c^2 \approx 26.7 \text{ MeV per helium-4}, \qquad L/c^2 \approx 4.3 \times 10^{9} \text{ kg s}^{-1}$$

3 / WAVE AND BLACKBODY / EMISSION AND THE JOURNEY, h WITH c AND k

The photosphere radiates a Planck spectrum, and the light takes 8.3 minutes

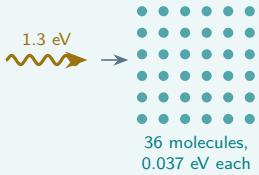


At $T = 5772$ K the spectrum depends on $hc/(\lambda kT)$ and peaks at $\lambda_{\max} = hc/(4.965\,kT) = 502$ nm, where a photon carries 2.5 eV; the mean photon energy is $2.70\,kT = 1.34$ eV. The power $L = 4\pi R^2 \sigma T^4 = 3.83 \times 10^{26}$ W uses $\sigma = 2\pi^5 k^4/(15h^3c^2)$, a law holding all three constants. Light covers $d/c = 499$ s to Earth and thins over $4\pi d^2$ to $S = 1361$ W m^{-2}, about 6×10^{21} photons per square metre per second.

$$S = \frac{L}{4\pi d^2} = 1361 \text{ W m}^{-2}, \qquad t = \frac{d}{c} = 8.3 \text{ min}$$

4 / ATOMIC AND THERMAL / ARRIVAL AND HEATING, k

An absorbed photon becomes the thermal motion of about 36 molecules



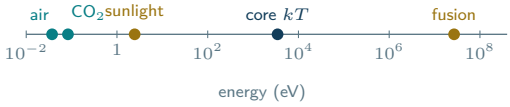
Sunlight is 30% reflected, and the average absorbed flux $S(1-0.30)/4 = 238$ W m^{-2} balances σT_e^4 at $T_e = 255$ K, whose spectrum peaks near 11 μm. Ozone takes the ultraviolet and water vapour and CO_2 the infrared, the 15 μm bend of CO_2 carrying 0.083 eV, and collisions convert that into motion with $\langle KE \rangle = \frac{3}{2}kT = 0.037$ eV per molecule at 288 K. One 1.3 eV photon thus warms about 36 molecules, each N_2 moving at $v_{\text{rms}} \approx 507$ m s^{-1}.

$$\langle KE \rangle = \tfrac{3}{2}kT = 0.037 \text{ eV at } 288 \text{ K}; \quad \text{one } 1.3 \text{ eV photon} \approx 36 \text{ molecules}$$

5 / SYNTHESIS

The same energy, from MeV to hundredths of an eV, through three conversion factors

stage	const.	relation	energy
fusion in the core	c	$\Delta m\, c^2$	26.7 MeV
core thermal photon	k	$2.7\,kT$	3.5 keV
sunlight at its peak	h	hc/λ	2.5 eV
CO_2 infrared photon	h	hc/λ	0.083 eV
air molecule	k	$\frac{3}{2}kT$	0.037 eV



ASIDE / THE TWO RADIATION LINKS

Accelerating charges and hot bodies both radiate, which is why the Sun's light is where the constants meet.

ASIDE / CHECK IN PYTHON

`2*pi**5*k**4/(15*h**3*c**2)` gives 5.670×10^{-8}, and `(1361*0.7/4/sigma)**0.25` gives 254.6, with h, c, k, σ from `scipy.constants`.

Convention. Figures are order-of-magnitude solar values ($T = 5772$ K, albedo 0.30); the diagram follows the constants map of mass, length and temperature.

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